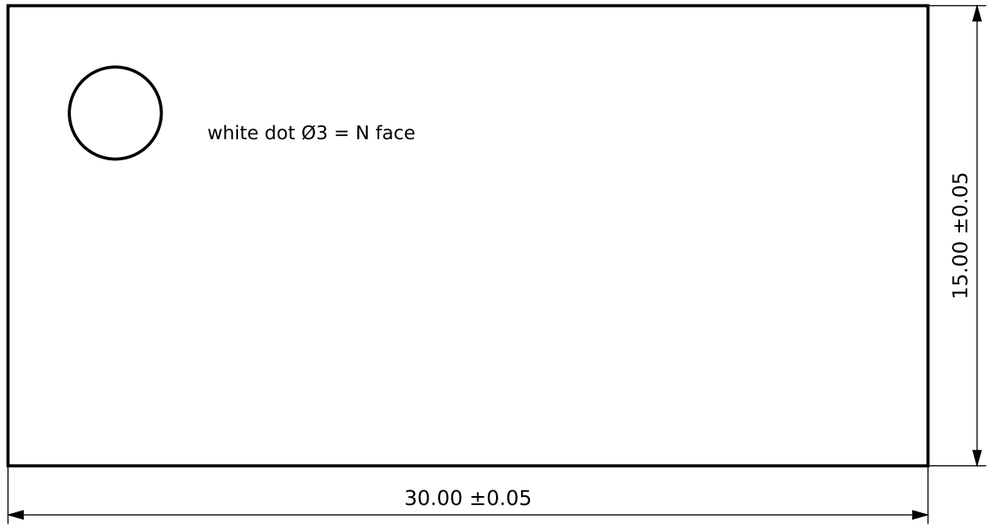
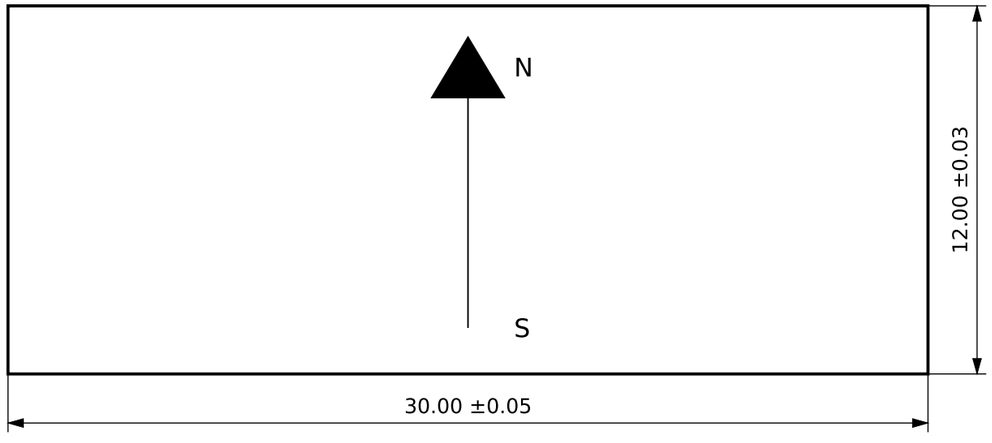
VIEW A: N FACE (5:1)



VIEW B: END, magnetised through 12 (5:1)



Item	Requirement
Grade	N45SH: Br ≥ 1.32 T, HcJ ≥ 1,592 kA/m (20 kOe), (BH)max ≥ 342 kJ/m³ at 20 °C
Curves	J-H and B-H per batch at 20, 60, 100, 120 °C; αBr, βHcJ stated
Magnetisation	Through the 12 mm thickness, saturated; axis within 2° of the face normal
Tolerances	30 ±0.05, 15 ±0.05, 12 ±0.03; faces flat 0.02, parallel 0.02
Edges	Chamfer 0.2 to 0.3 × 45° all 12 edges; no chips > 0.3
Coating	NiCuNi 10 to 20 µm; 72 h neutral salt spray clean; adhesion OK with the epoxy
Uniformity	Moment of each piece within ±1.5 % of the lot mean; value per serial
Marking	White dot Ø3 on the N face; serial on the packaging position
Quantity	200 + 10 spare per generator; 8 extra for Q1 tests
Packing	Repelling layout in non-magnetic spacers; air freight per IATA PI 953

NOTES

1. One part for N and S poles: N poles face N towards the stator, S poles are the same block flipped (dot hidden against the seat).
2. Solid block (not two 6 mm pieces bonded). If a maker offers only 30 × 15 × 6, the doc 20 bonded pair is the fall-back with identical magnetics.
3. Five blocks per pole along the stack (15 mm pieces = axial segmentation for eddy loss, 4.7 W at rated).
4. Release of N45SH for production is conditional on the 100 °C design basis (doc 19 §16.3, doc 23 §4): bench magnet temperature ≤ 75 °C, temperature watch, no deliberate phase shorting.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

MAGNET BLOCK 30 × 15 × 12 (PURCHASE DRAWING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-220E	E · 2026-11-05	5:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
Sintered NdFeB N45SH, NiCuNi	41.0 g each, 8.21 kg per generator	Claude / not checked